Yongkuk Jeong, PhD

Assistant Professor in Sustainable Production Logistics
Department of Production Engineering, KTH Royal Institute of Technology
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Work Experience

KTH Royal Institute of Technology, Sweden — Jan 2021 – Present
Assistant Professor, Department of Production Engineering

Pohang University of Science and Technology (POSTECH), South Korea — Jan 2026 – Dec 2027
Affiliate Faculty, Department of Industrial & Management Engineering (concurrent appointment)

KTH Royal Institute of Technology, Sweden — Jan 2019 – Jan 2021
Postdoc, Department of Sustainable Production Development

Inha Technical College, South Korea — Mar 2018 – Dec 2018
Lecturer, Department of Naval Architecture and Marine Engineering

Seoul National University, South Korea — Mar 2018 – Jan 2019
Postdoc, Research Institute of Marine Systems Engineering

Education

Seoul National University, South Korea — Sep 2011 – Feb 2018
Doctor of Philosophy in Naval Architecture and Ocean Engineering
- *Thesis title: A shipyard logistics simulation system considering shipbuilding process, spatial arrangement, and logistics flow*

Seoul National University, South Korea — Mar 2007 – Feb 2011
Bachelor of Science in Naval Architecture and Ocean Engineering
- Graduated as valedictorian in the program

Selected Publications

[1] Enabling Industrial Internet of Things-based Digital Servitization in Smart Production Logistics.
- Erik Flores-García, *Yongkuk Jeong*, Sichao Liu, Magnus Wiktorsson, and Lihui Wang (2022).
- International Journal of Production Research, 61 (12).

[2] Spatial Arrangement using Deep Reinforcement Learning to Minimise Rearrangement in Ship Block Stockyards.
- Byeongseop Kim, *Yongkuk Jeong*, and Jong Gye Shin (2020).
- International Journal of Production Research, 58 (16).

[3] A Simulation-based Decision Support Tool for Circular Manufacturing Systems in the Automotive Industry using Electric Machines as a Remanufacturing Case Study.
- Mayarí Pérez, Michael Lieder, *Yongkuk Jeong,* and Farazee M. A. Asif (2025).
- International Journal of Production Research.

[4] Machine Learning in Smart Production Logistics: A Review of Technological Capabilities.
- Erik Flores-García, Dong Hoon Kwak, *Yongkuk Jeong*, and Magnus Wiktorsson (2025).
- International Journal of Production Research, 63 (5).

[5] A Framework and System Architecture for Value-oriented Digital Services in Data-driven Production Logistics.
- Masoud Zafarzadeh, Erik Flores-García, *Yongkuk Jeong*, and Magnus Wiktorsson (2025).
- International Journal of Production Research.

[6] A Review on Digital Twin Application in Integrated Ship Design and Production for Decarbonisation.
- Chao Ye, Abbas Dashtimanesh, *Yongkuk Jeong*, and Hauwa Yunusa (2026).
- Ocean Engineering, 359.

[7] AI-enabled Vision Systems for Human-centered Order Picking.
- Erik Flores-García, Enrique Ruiz Zúñiga, *Yongkuk Jeong*, and Magnus Wiktorsson (2026).
- International Journal of Production Research, 64 (5).

More publications are available in Google Scholar profile and my personal website

Selected Research Projects

ALCHEMIST Jan 2025 – Dec 2029

A human-centric manufacturing multiverse with AI avatars (Smart Manufacturing Multiverse Platform based on Multisensory Fusion Avatar and Interactive AI)
- Funded by the KEIT Industrial Technology Alchemist Programme (South Korea)
- Work package leader for the KTH team, on a secure GenAI- and digital-twin-based multiverse platform for production logistics

cLOK From 2025

Conversational AI for production planning (Interactive AI Scheduling Platform for Human-centric Industry 5.0)
- Funded by the National Research Foundation of Korea (NRF), mid-career research programme
- International collaborator, working with Seoul National University

Data-driven production logistics with AstraZeneca Aug 2026 – Jan 2027

Exploration phase for digitalizing and automating internal production logistics in pharmaceutical manufacturing
- Funded by Digital Futures (ISPP Mobility Grant), with AstraZeneca, Södertälje
- Applicant and lead

SHIFT-DT Jan 2024 – Dec 2025

Sustainable, Holistic, Integrated Framework for Ship Design and Production Transformation through Digital Twins
- Funded by Digital Futures
- Leading the project as a Principal Investigator (PI) to establish a framework that can marry holistic ship design with digitalized ship production and logistics through digital twins

Dynamic SALSA Apr 2023 – Mar 2026

Dynamic scheduling of assembly and logistics system using AI
- Funded by Vinnova (Eureka SMART)
- Building a computer vision-based platform for analyzing human operators and objects in a production logistics environment

TIMEBLY & TIMEBLY 2 Nov 2021 – Oct 2024

Time data management automation for manual assembly — Sep 2025 – Aug 2028
- Funded by Vinnova
- Leading the human pose estimation and time series prediction analysis for manual assembly tasks using open-source libraries

DYNASTEEL — Mar 2022 – Dec 2022
Dynamic scheduling and transport visibility in steel production
- Funded by Vinnova
- Involved in human-centered system design process for autonomous transportation system including requirements analysis and prototyping

C-PALS — May 2019 – Aug 2022
Cyber-physical assembly and logistics system
- Funded by Vinnova (Eureka SMART)
- Built a digital twin and real-time production logistics data visualization platform using open-source libraries

HUPMOBILE — Jan 2019 – Dec 2021
Holistic urban and per-urban mobility
- Funded by Interreg Baltic Sea Region
- Involved in participatory modelling and simulation process for urban mobility simulation with multiple stakeholders

Simulation system for manufacturing strategy and execution to quantify ship production cost — Oct 2016 – Jan 2019
- Funded by National IT Industry Promotion Agency of Korea
- Developed and implied a simulation-based monitoring system for ship production process

Simulation-based production planning and management system for middle-sized shipbuilding companies — Oct 2014 – Jan 2019
- Funded by Ministry of Trade, Industry, and Energy of Korea
- Developed and implied an advanced planning and control system for ship production process

Advanced Naval Vessel Research Laboratory — Sep 2012 – Dec 2017
- Funded by Ministry of National Defense of Korea
- Developed an optimization algorithm for spatial layout design of naval vessels considering the survivability

Teaching Experience

KTH ML2302: Modelling, Simulation and Optimization of Sustainable Production — 2020 – Present
Course responsible and teacher

KTH ML2303: Digitalisation for Sustainable Production — 2020 – Present
Course responsible and teacher

KTH ML2305: Production Logistics and Supply Chains — 2022 – Present
Teacher and Examiner (2024 – present)

KTH ML230X: Degree Project in Sustainable Production Development, Second Cycle — 2020 – Present
Supervisor

KTH ML1503: Industrial Systems II — 2020 – Present

Guest lecture in Industry 4.0

KTH ML1505: Industrial Systems III 2020 – Present
Guest lecture in Operator and Industry 4.0

KTH ML2307: Theory of Science and Research Methodology in Sustainable Production Development 2022 – 2024
Teacher and course responsible (2022 – 2024)

Ingenjör4.0 - Upskilling for future manufacturing 2022 – 2024
Course responsible for Autonomous Robots and Cyber-Physical Systems

EIT Urban Mobility Lifelong Learning in New Trends on Urban Mobility 2022
Teacher

InhaTech: Ship Production Design 2018
Course responsible and teacher

InhaTech: Introduction of Ship Production Engineering 2018
Course responsible and teacher

Skills

- **Project management**: led and successfully delivered various research and development projects
- **Programming skills**: C#, Python, JavaScript, Java, SQL, HTML, and UML
- **Language**: English (fluent), Korean (fluent), Swedish (working knowledge)

Others

- **Winner of the Elmer L. Hann Award** for best paper on Ship Production delivered at a Society of Naval Architects and Marine Engineers (SNAME) event in 2019 for the paper "Model-based Computational Shipyard Dynamics and its Applications"
- **Reviewed papers** for various journals and conferences including:
  - International Journal of Computer Integrated Manufacturing (IJCIM)
  - International Journal of Production Research (IJPR)
  - Ships and Offshore Structures
  - Automation in Construction
  - Journal of Engineering for the Maritime Environment (JEME)
  - International Journal of Naval Architecture and Ocean Engineering (IJNAOE)
  - Journal of Ship Production and Design (JSPD)
  - International Journal of Precision Engineering and Manufacturing-Green Technology (IJPEM-GT)
  - IFIP International Conference on Advances in Production Management Systems (APMS)
  - Winter Simulation Conference (WSC)
  - European Operations Management Association (EurOMA) annual conference
- **Editorial board member** in
  - International Journal of Sustainable Engineering
  - Ships and Offshore Structures
  - International Journal of Precision Engineering and Manufacturing – Smart Technology
- **Life member** of the Society of Naval Architects of Korea